Introduction to Artificial Intelligence

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Lecture 1: Introduction to AI

1.1 Rational decisions

AI can be framed as making machines think or act like people, or think or act rationally. This course focuses on RATIONAL ACTION: choosing an action that best achieves a specified goal given the available information.

UTILITY $U(o)$ measures the desirability of outcome o. Given information h, an action a, and possible outcomes O, choose an action that maximizes expected utility:

$$a^* \in \arg\max_a \sum_{o \in O} P(o \mid h, a) U(o).$$

Probabilities describe uncertainty; utility describes the goal. Rationality evaluates the decision using information available at the time, so a rational choice can still have a bad outcome.

Example: A safe route has utility 6. A shortcut has utility 10 with probability 0.7 and -5 otherwise. Its expected utility is $0.7(10) + 0.3(-5) = 5.5$, so the safe route is preferred. If the success probability is p, the shortcut is better when $15p - 5 > 6$, or $p > 11/15$.

1.2 Agents and environments

An AGENT receives PERCEPTS through sensors and acts on an environment through actuators. The environment changes and produces further percepts.

- A REFLEX AGENT selects an action from its current percept or internal state without explicitly evaluating future outcomes.
- A PLANNING AGENT uses a model to compare possible future consequences before acting. A plan may need revision when new observations arrive.

Example: In Pac-Man, observations may include walls, food, and ghosts; actions choose a direction. A planner can compare routes to food while accounting for ghosts. If the model assumes ghosts never move, the best route in that model may not be safe in the actual game.

The task determines what state matters: a route to a location may require position and legal moves, while collecting all food also requires the remaining food locations. Limited observations may require memory or a belief about hidden state.

1.3 Methods and applications

- **Search and planning** compare action sequences; **constraint satisfaction** finds assignments obeying rules; game search accounts for other decision makers.
- **Probability** represents uncertainty; **decision theory** combines uncertainty with goals; **machine learning** uses data to improve predictions or decisions.
- Applications include language, vision, robotics, and games. A system often combines several methods: for example, estimates of traffic from data can feed a route planner.

AI has progressed through symbolic search and reasoning, knowledge-based systems, statistical approaches, and systems using large datasets, computation, and neural networks. Different tasks still require appropriate representations and objectives.

Source. Pieter Abbeel and Dan Klein, Introduction to Artificial Intelligence, Fall 2018, Lecture 1 and Note 1.